
Usability Evaluation Presentation

GROUP 15

F416946

Introduction

Louis Vuitton website



A brief statement

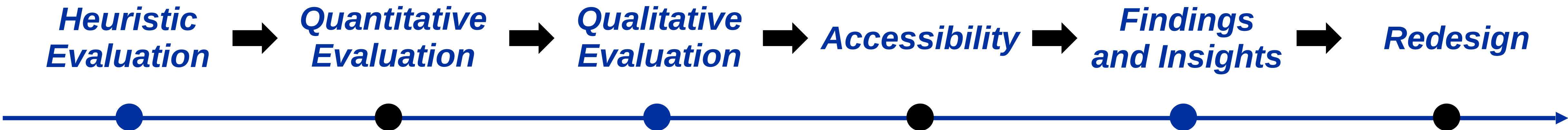
- An online luxury retail platform.
 - A variety of high-end products
-

The aim of the evaluation

- Evaluate the usability of the website
- understand how well the website works for users
- looking for any problems that could make it hard for people to navigate or complete tasks.
- find ways to improve the website and make it easier for users to shop
- accessibility, ensuring that the website is inclusive and usable for all people, including those with disabilities.

Contents

1 Reporting Process



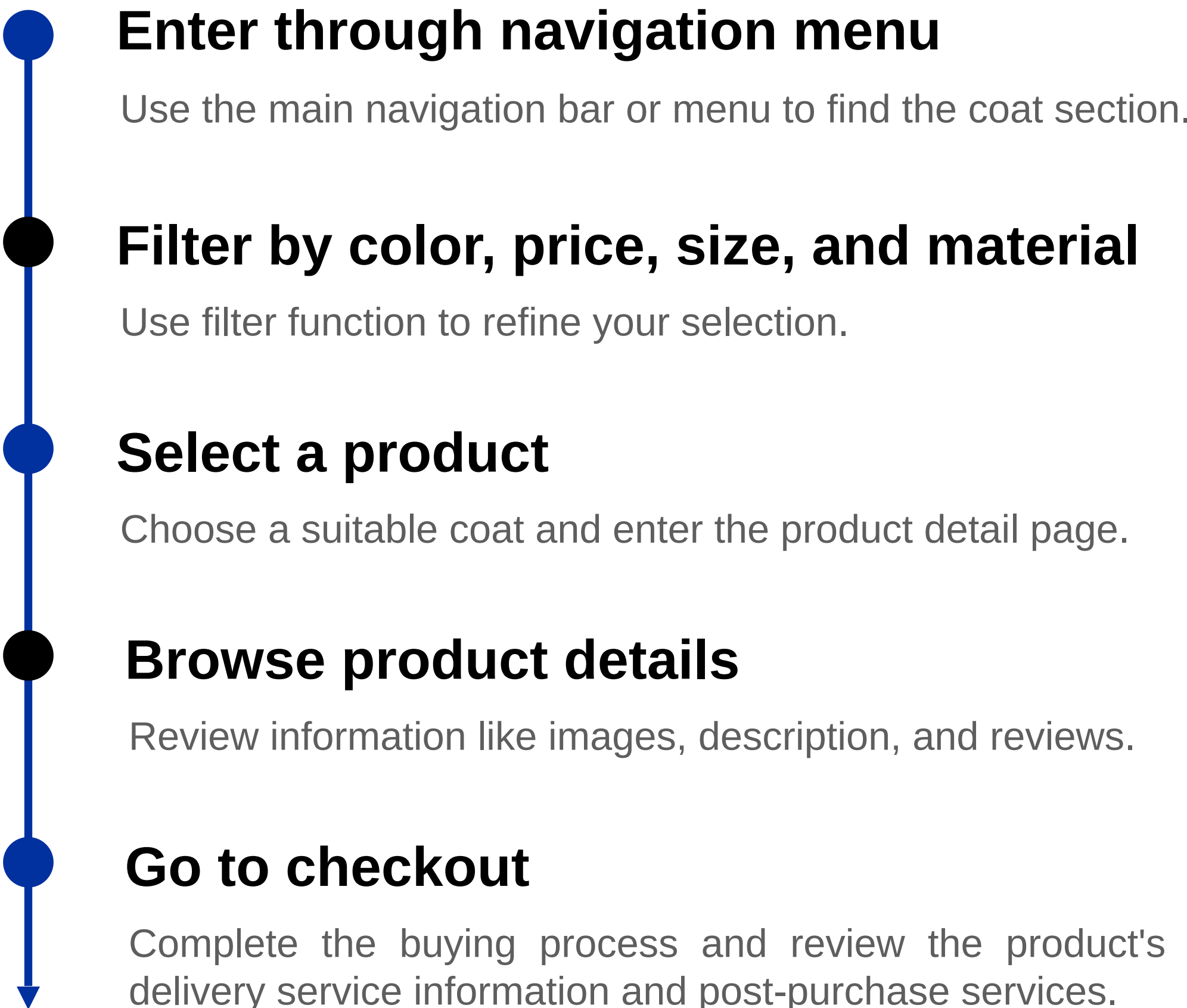
2 Overview of Methodology

Heuristic Evaluation	Quantitative	Qualitative	Accessibility	Findings and Insights	Reflection
• Usability Principles • Guidelines: Nielsen's 10 Usability Heuristics • Mental Models: Aligning interface design with user expectations • Task Completion: Identifying navigational issues and usability barriers	• Measure specific metrics to understand how well the website performs • Task Time • Task Success Rate • Error Rate • SUS Score • Lostness	• Gather insights from real users. • Observed the interactions and asked about their experiences • User Feedback • Usability Issues • User Expectations • Interviews and Observations	• Ensures the website is usable for people with disabilities • WCAG Guidelines • Text Readability • Screen Reader Compatibility • Keyboard Navigation • Color Contrast • Alternative Text	• Usability Issues • User Behavior • Pain Points: Areas where users experience frustration or difficulty • Key Points: Insights into user expectations and problems encountered	• Critically thinking • What worked well • Challenges faced • Lessons learned • Recommendations for future evaluations

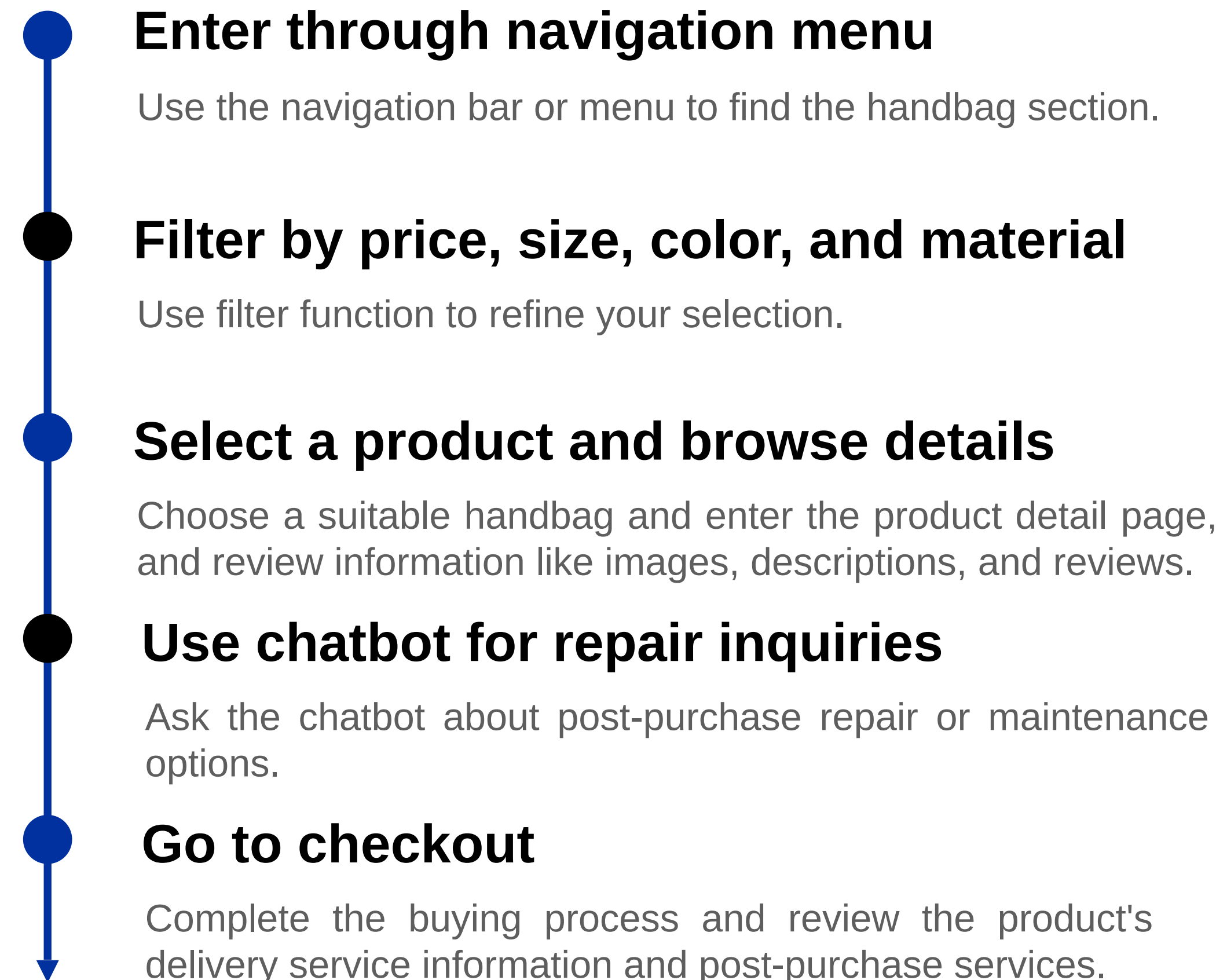
Heuristics Evaluation Work Plan

Goal: Find a brown coat and a handbag through the navigation bar and filters, and buy it. (Individual)

1 *Work plan 1*

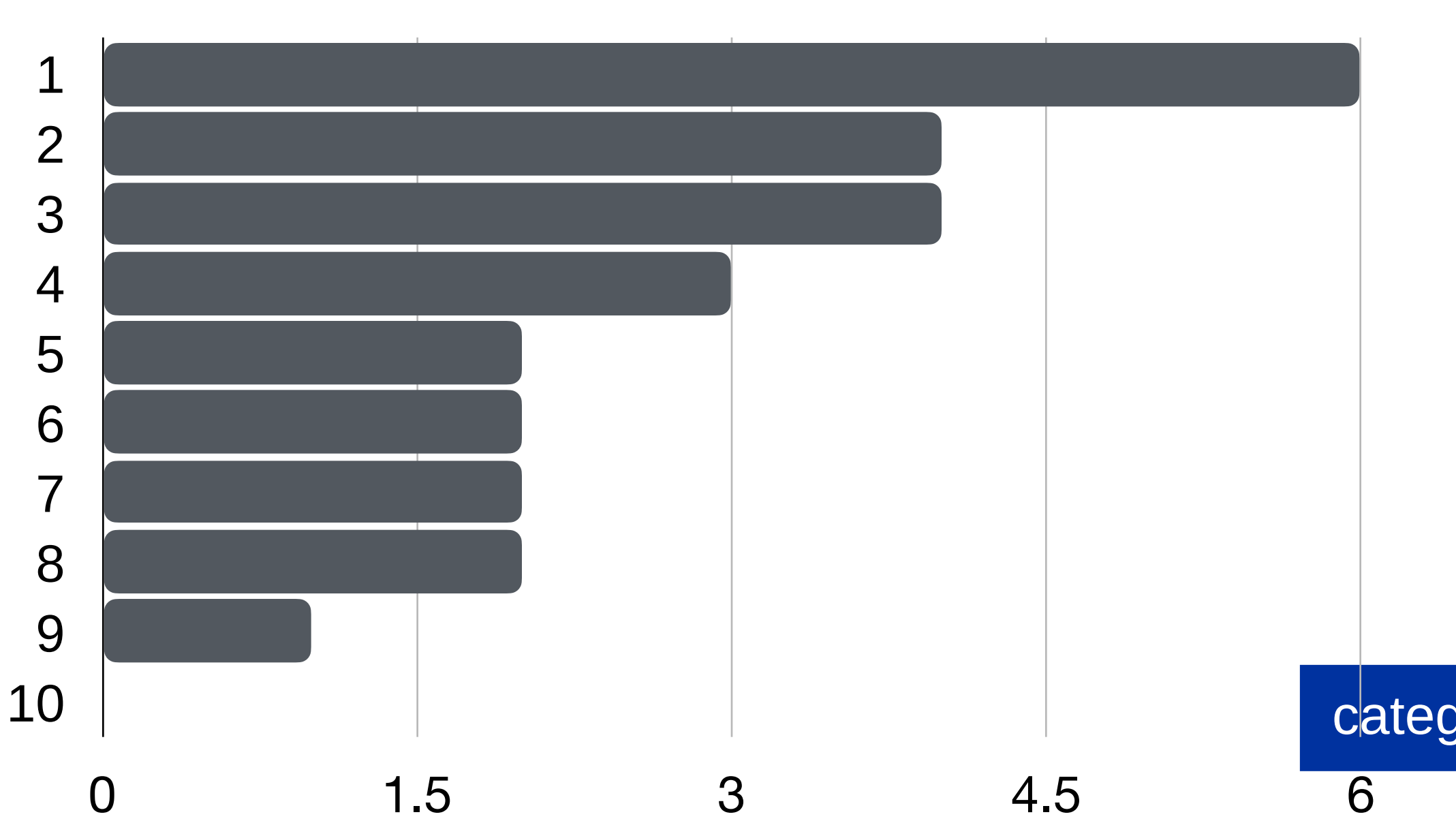


2 *Work plan 2*



Heuristic Evaluation

Here 26 Usability Issues



1. Consistency and standards

2. Recognition rather than recall

3. Flexibility and efficiency of use
8. Aesthetic and minimalist design

9. User control and freedom

10. Error prevention
4. Match between system and the real world

5. Visibility of system status

6. Help and documentation

7. Help users recognize, diagnose, and recover from errors

categorise

Here 8 tasks

	Usability catastrophe	Major problems	Minor problems	Cosmetic problems
	Usability catastrophe	Major problems	Usability catastrophe	Usability catastrophe
Filtering and menu				
Search				
Product description				
Check out				
Returns				
Delivery				
Customer reviews				
Image				

Quantitative Tasks

Goal: Find 'Sunglasses' using Filters, gather information about product care and proceed to the Checkout section. Find important information related to Delivery and Payment before buying it.

Tasks

Start from the Home Page

1

Minimum time = 65s
R(for Lostness)= 11

- Begin at the website's homepage.
- Click on **Menu**, and select **Men** or **Women**, then choose **Accessories**.
- Click **Sunglasses** under Accessories.
- Filter by **Shape**: Square.Filter by **Color**: Black.

Chosen Product Page

2

Minimum time = 36s
R(for Lostness)= 7

- Click the **first product** on the page.
- Check the **Transmittance** in the product details.
- Go back and click the **second product**, and check the **Transmittance** again.
- Find and click the **first product** in your recently viewed list.
- Locate **care instructions** for the sunglasses.
- Add the product to your **Wishlist**.

Shopping Bag Page

3

Minimum time = 17s
R(for Lostness)= 5

- Add to Shopping Bag and go to Checkout.
- Find the quickest delivery time option and Check for **UK addresses** with delivery restrictions.
- Find out the **processing time** for a bank transfer payment.
- Check the nearest **store availability** for the sunglasses.
- Go to the Shopping Bag, **change quantity to 2**.
- Tell the researcher when finished.

Quantitative Data

Task 1

Difficulty Finding Product Care Information

Heuristic Principle:

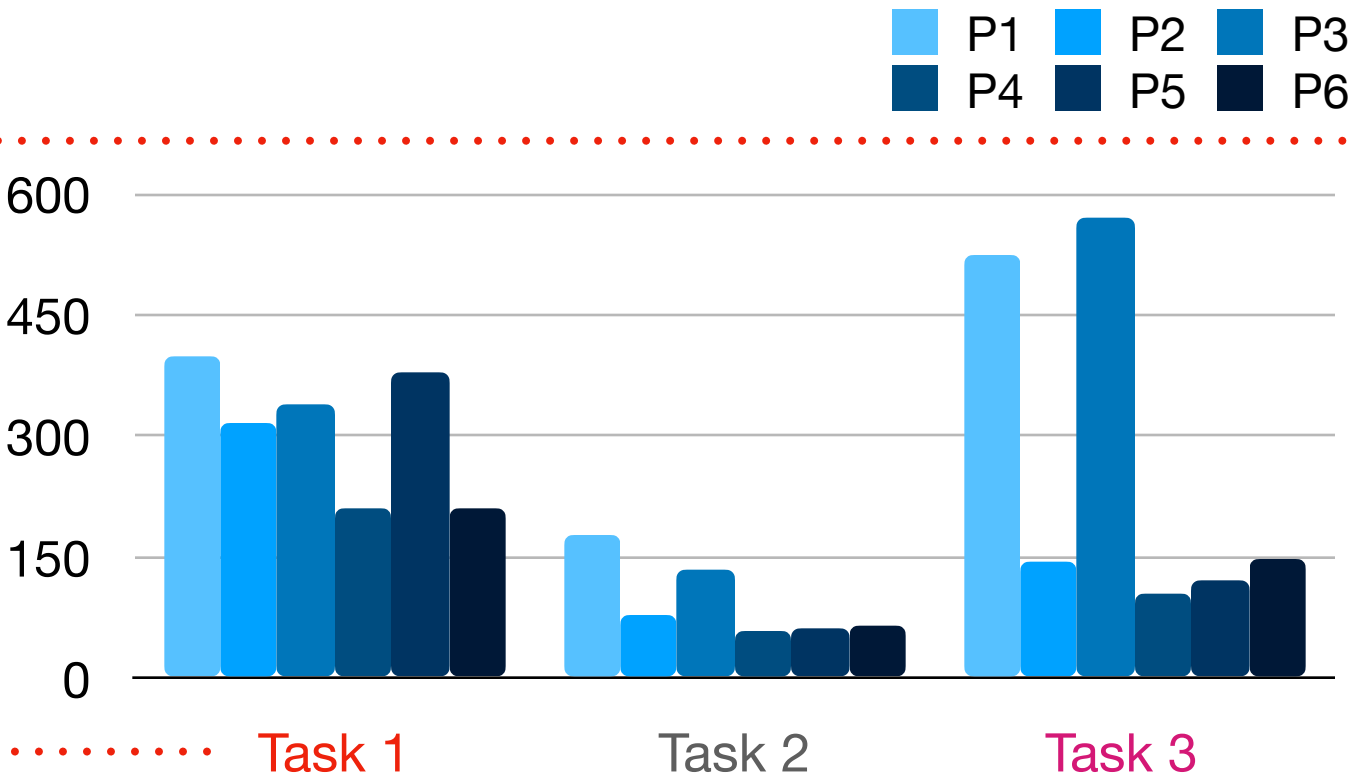
Visibility of System Status: Important information like "Product Care" should be more visible, not in a small font at the bottom of the page.

Recognition over Recall: The location of "Product Care" requires memorization or extensive searching and is not easily identifiable and found

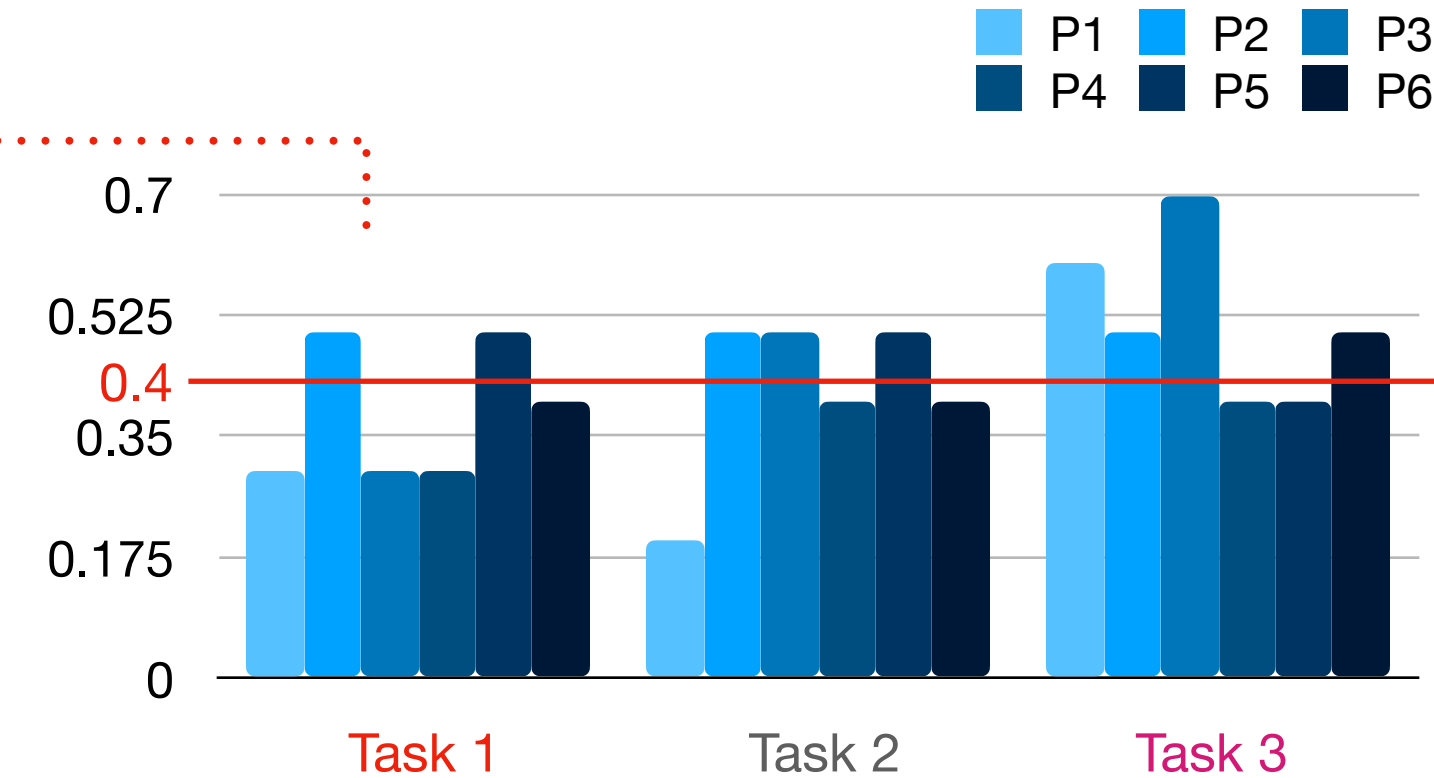
Psychology Principle:

Fitts' Law: Small font and bottom placement make it hard to access and click.

1 Task time

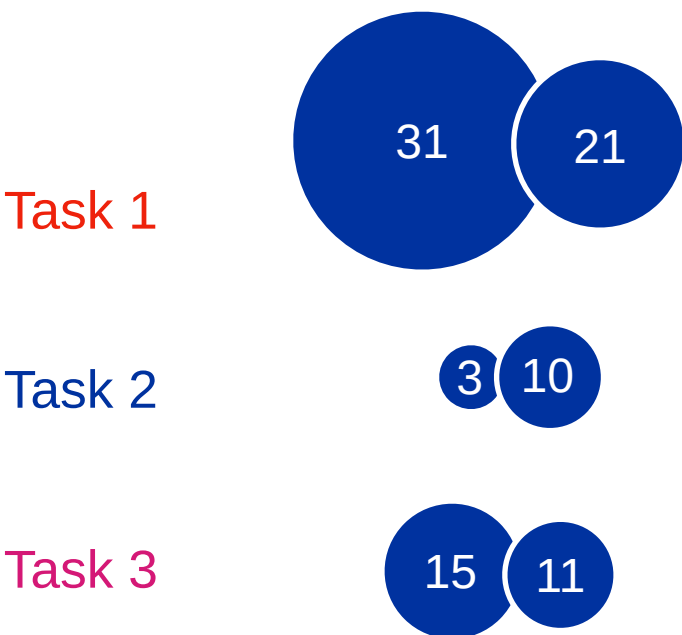


2 Lostness

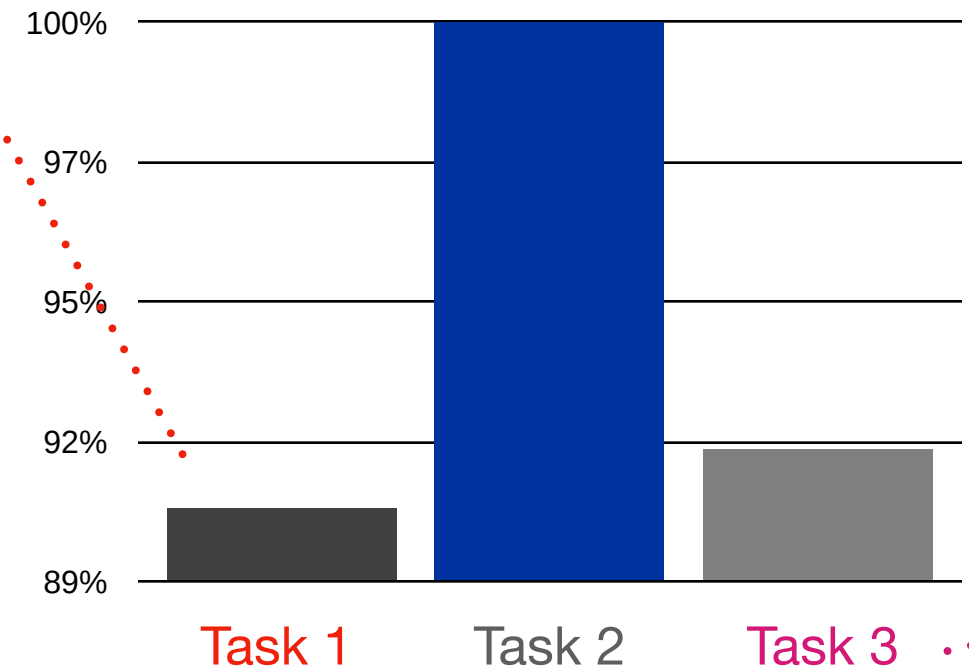


Average Lostness	
Task 1	0.4
Task 2	0.3
Task 3	0.5

3 Errors made



4 Task completion



Task 3

Difficulty Reading and Understanding Delivery Information

Heuristic Principle: Psychology Principle:

Heuristic Principle:

Help Users Recognize, Diagnose, and Recover from Errors: Users struggled with complex text. Especially new and international user.

Psychology Principle:

Mental Models: Complex wording and dense text increase cognitive load.

Reporting SUS score

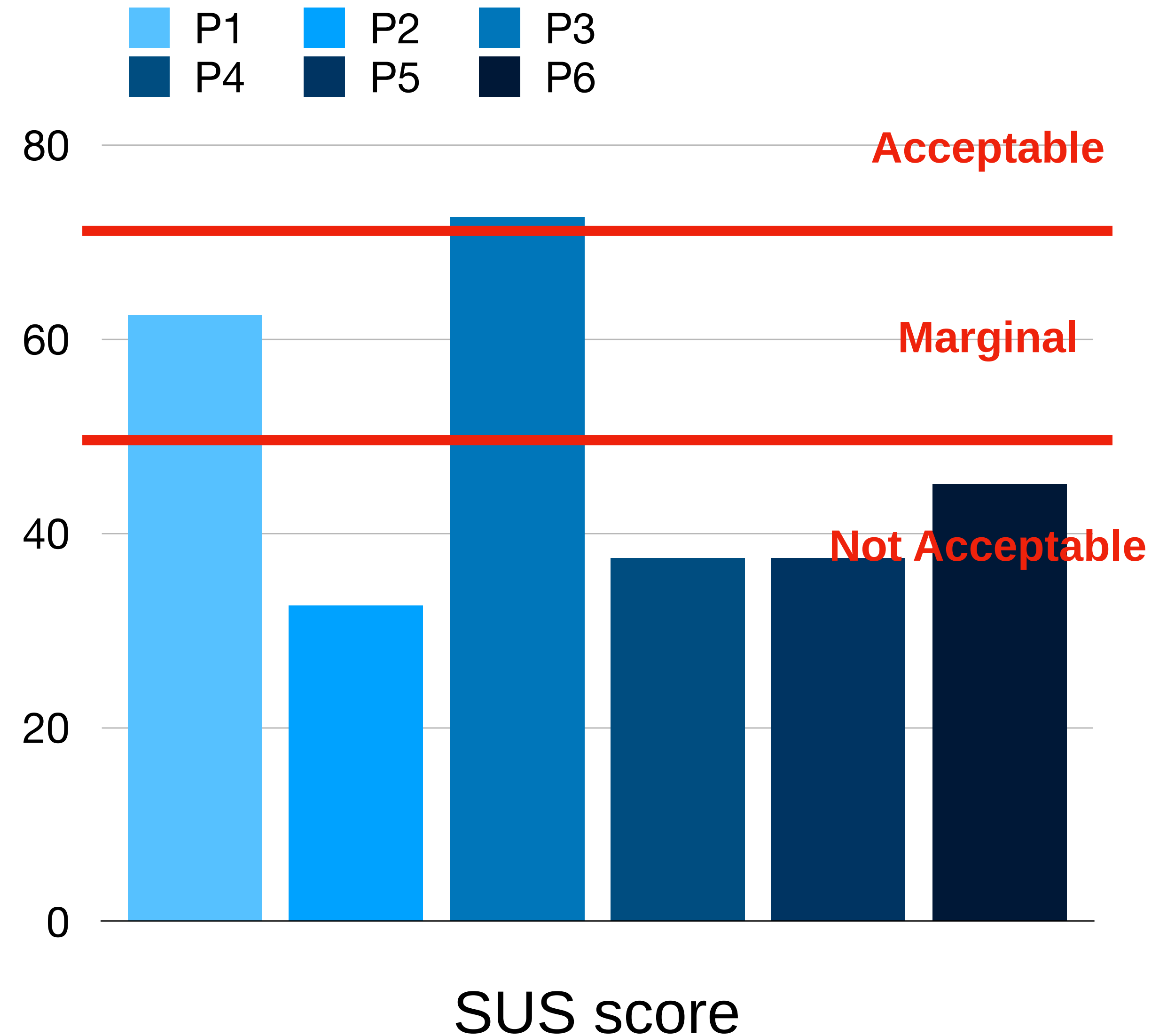
SUS Grade

- ***< 50 = not acceptable***
- ***50 to 70 = marginal***
- ***70 >= acceptable***

SUS Score Analysis

Positive and Negative Problem:

- Higher scores on negative statements like “**unnecessarily complex**” and “**very cumbersome to use**” —suggest usability challenges.
- Lower scores like “I would imagine that most people would learn to use the website very quickly” highlight areas for improvement.



Qualitative Tasks

► Scenario:

You have arranged a vacation to Iceland for Christmas trip, and you need to buy a new travel luggage and a female/male coat for the cold winter weather for yourself.

► Mission:

Find the most stylish travel luggage within the £2,000-4,000 range and a coat within the £3,000-5,000 range. Check that the luggage size meets airline cabin allowances and select the correct coat size. Review the product care details for each item, then add both to your shopping bag and proceed to checkout.

Qualitative Findings

	WHAT THEY DO	WHAT THEY SAY	WHAT THEY THINK
<u>Travel Luggage</u> • Effectiveness: 80% • Efficiency: Low • Satisfaction: Low	Luggage • Use filters frequently to search by price • Repeatedly check size details to ensure suitability for airline requirements	Luggage • "I don't know if 'cabin size' means it's carry-on." • "The filter doesn't show available products." • "I'm not familiar with terms like 'keepall' and 'Horizon.'"	Luggage • Confused: Size and availability info is unclear, making it hard to find suitable options. • Frustrated: Filtering doesn't reliably display available items, and stock information is unclear. • Pleased: Some filters (like price) work well, offering a pleasant surprise.
<u>Coat</u> • Effectiveness: 80% • Efficiency: Low • Satisfaction: Low	Coat • Use the size chart to aid size selection. • Repeatedly check inventory status to confirm product availability. • Attempt price and size filtering but find some options unavailable.	Coat • "The size chart is surprisingly helpful." • "Products show in stock but are unavailable when selected." • "Why display options if they aren't functional?" • "I feel like giving up on this task."	Coat • Confused: Stock info is inconsistent, showing items available but out of stock when selected. • Satisfied: Finds the size chart useful for choosing the right fit.
<u>Product care</u> • Effectiveness: 20% • Efficiency: Low • Satisfaction: Very low	Product care • Repeatedly search the bottom navigation and product details for product care info but often fail to find it.	• "I can't find any product care info." • "Product care should be in the product details, not at the bottom of the page." • "The font is so small—it's hard to notice."	• Confused: Product care information is not included in the product details. • Frustrated: Difficult to find product care due to small font size, leading to feelings of helplessness.

Qualitative Findings

Font Size:

Participants found that small fonts made reading and navigation difficult,

- This violated the **accessibility heuristics** in the WCAG guidelines.

Product pages:

Participants found that unclear navigation and missing product information led to usability issues on product pages.

- This violated the **match between system and real world**
- Lack of clear product information leads to **confusion and reduces task efficiency**.

Price filters:

Participants found the price range filter difficult to use

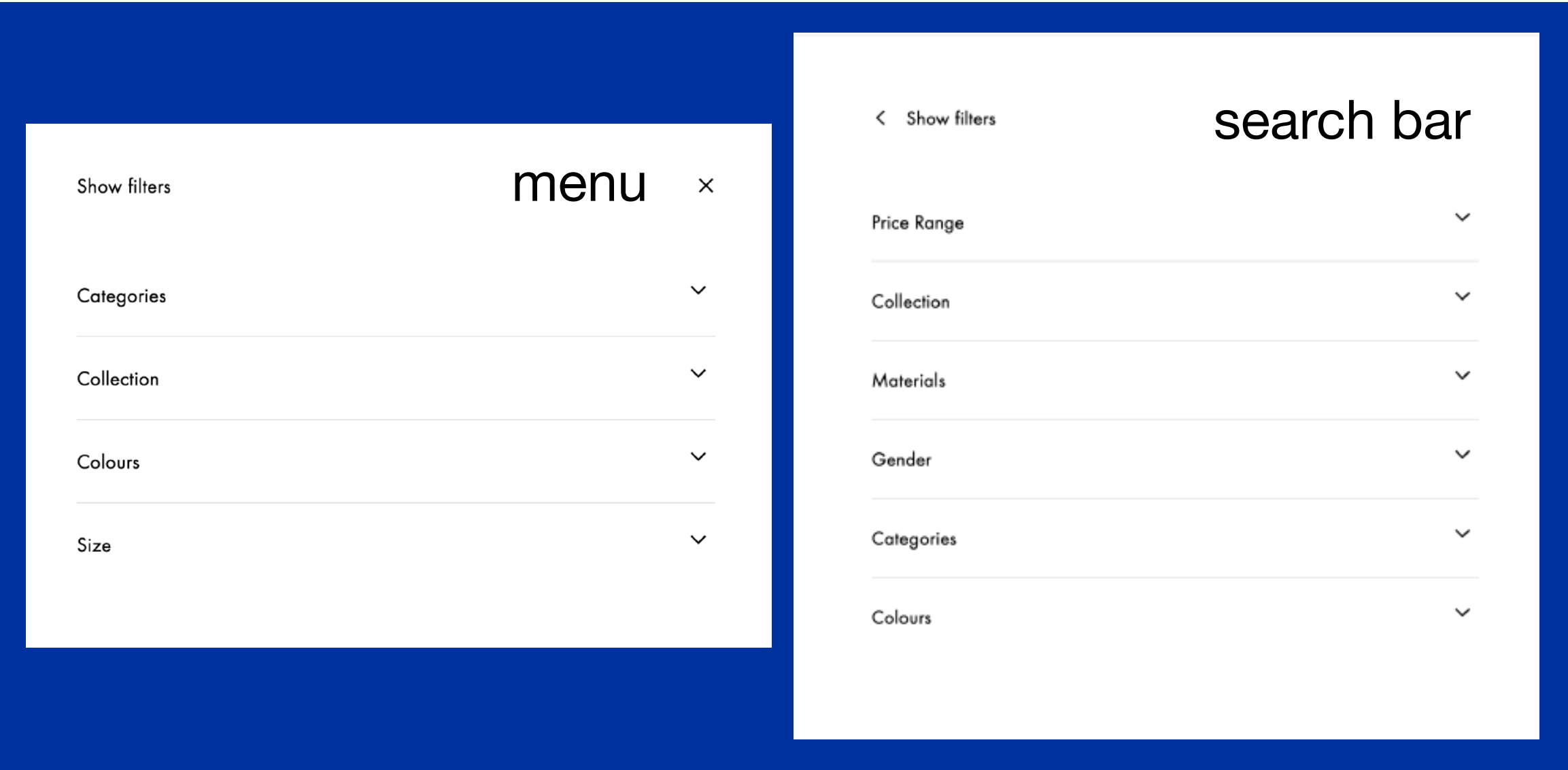
- This violated **consistency and standards**.
- Most e-commerce sites offer standardized filter, but LV website led to user frustration.

Product care:

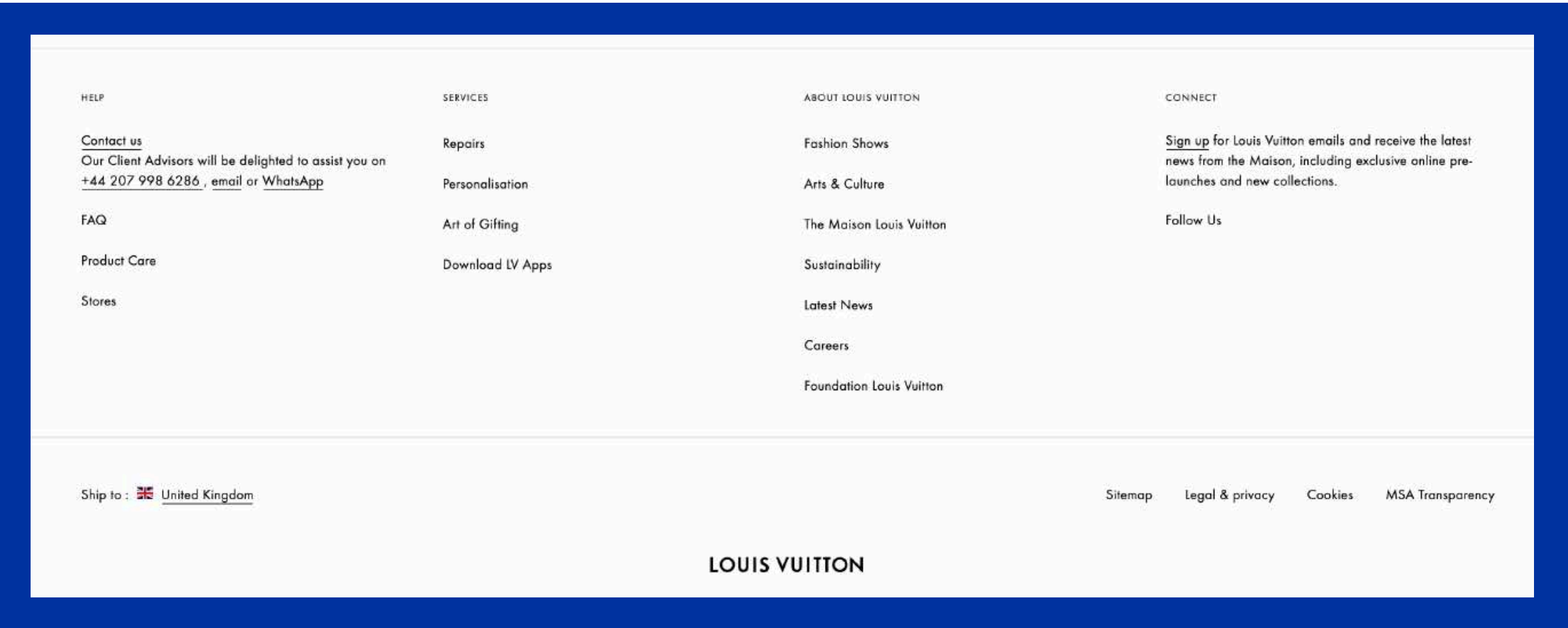
Participants had difficulty finding or understanding product care.

- This violated **recognition rather than recall**.

The same category of products (coat), different entrances show different filters.



Font size is too small.



Accessibility

“after conducting the accessibility evaluation I identified “35” accessibility violations. 29 were related to missing Alt text, 3 involved problems navigating with the keyboard and 3 were associated with contrast errors. In total 28 accessibility violations were perceivable and 4 we Operable”

Perceivable

Image missing alternative text

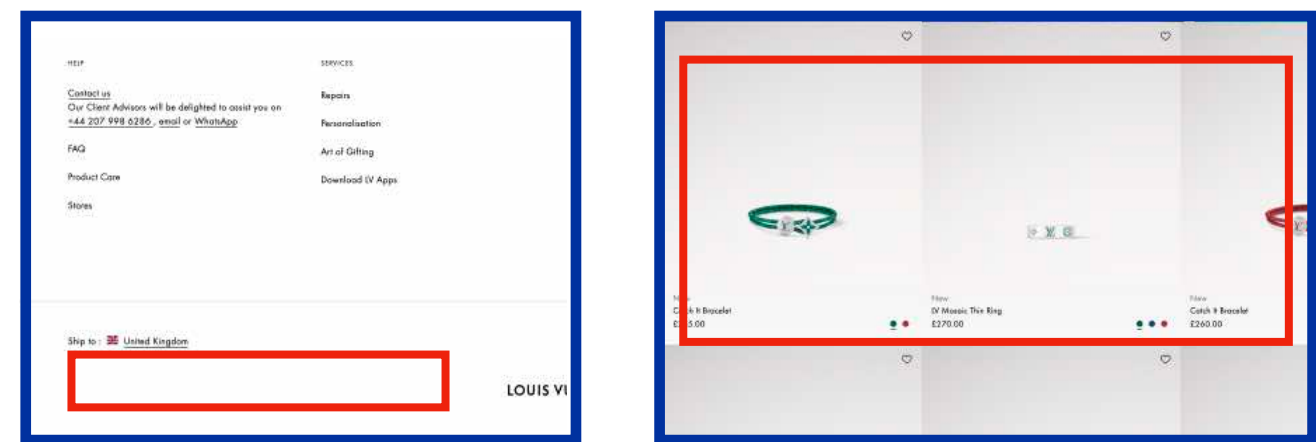
All images should have alternative text, and the space between the text and the image does not contain content or useful information

WCAG 2.1-1.1.1 Non-text Content(Level A)

Linked image missing alternative text

WCAG 2.1-1.1.1 Non-text Content(Level A)

2.4.4 Link Purpose (In Context) (Level A)



low contrast between text and background colors

Text with too low a contrast ratio can be difficult to read for users with low vision, color blindness, and the elderly.

WCAG 2.1-1.4.3 Contrast (Minimum)(Level AA)

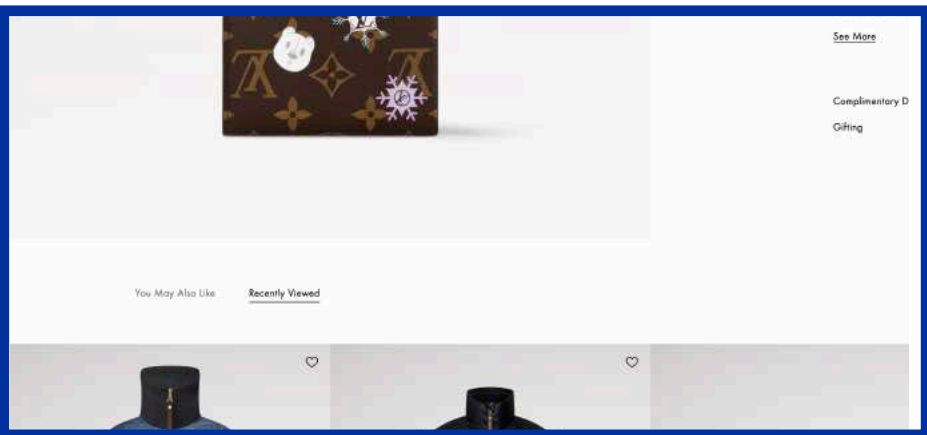


Operable

Hard-to-Find Recently Viewed Products

Recently viewed products are challenging to find, requiring multiple steps, which may not meet the requirement for easily accessible navigation paths.

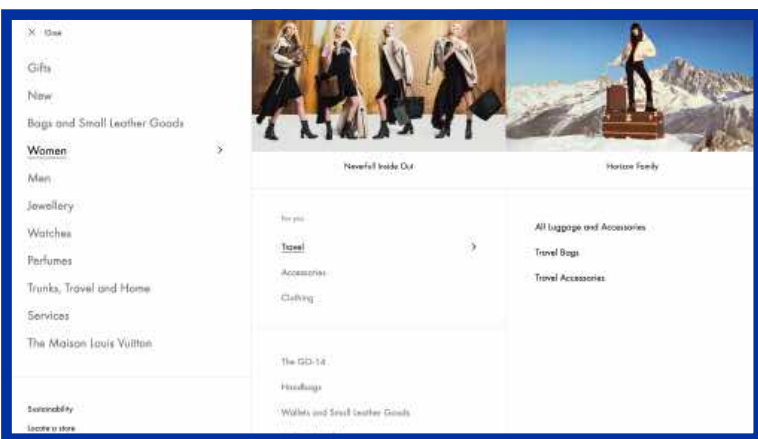
WCAG 2.1- 2.4.5 Multiple Ways (Level AA)



Poor Navigation Due to Overly Complex Menu

Excessive or repetitive menu options without clear structure can make navigation difficult, especially for users with cognitive impairments.

WCAG 2.1- 2.4.6 Headings and Labels (Level AA)



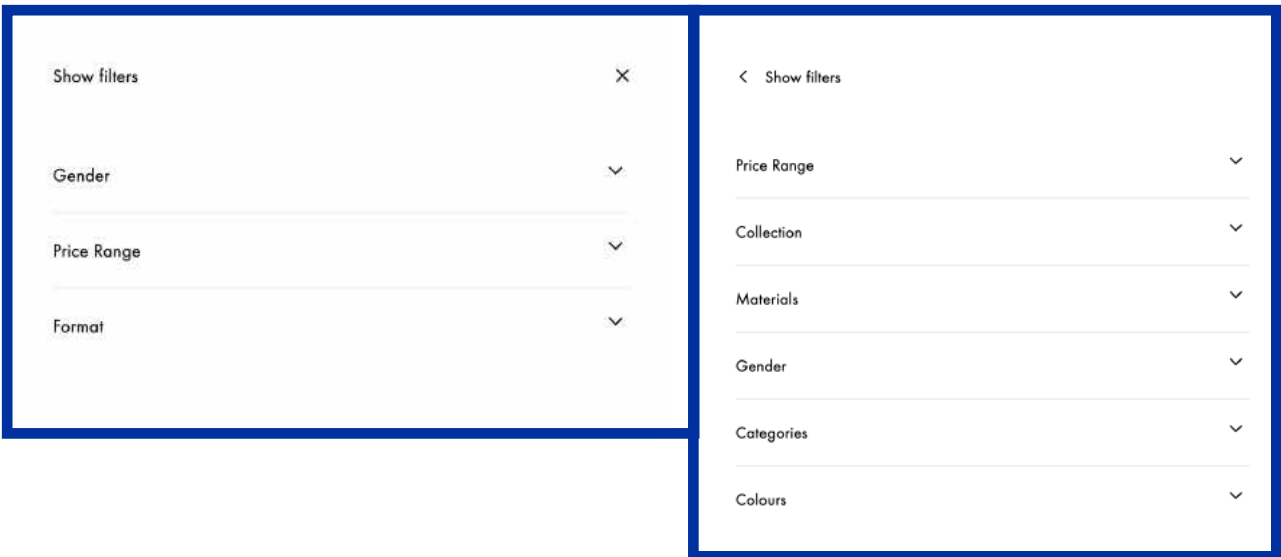
Understandable

Inconsistent Filter Function

Inconsistent filter options across pages, such as missing or non-standardized price filters.

WCAG 2.1- 3.2.4 Consistent Identification (Level AA)

3.2.3 Consistent Navigation(Level AA)



unusual words

Use complex words or jargon in product titles, with explanations in some places but not in others

WCAG 2.1- 3.1.3 Unusual Words(Level AAA)

3.1.5 Reading Level(Level AAA)



Reflection

Challenges & limitations

Heuristic

- As beginners in evaluation, we **missed some usability issues and could be subjective** in assessing the severity of problems.
- We **didn't fully consider user diversity**, including cultural backgrounds and disabilities. This might affect how different users understand the design and accessibility features.

Quantitative

- The website updates quickly, so products marked as available may be out of stock by the time of testing, leading to discrepancies.

Qualitative

- Testers had more freedom to perform tasks, which made it harder to compare results across tests.
- **Personal preferences or prior knowledge of testers affected their responses**, making it hard to generalize.

Accessibility

- **Lack of experience with WCAG guidelines** made it **harder** to fully assess the website's accessibility.
- Automated tools provided **useful but hard-to-interpret data**, making it challenging to evaluate accessibility properly.

To be improved

- *Include more diverse user perspectives, especially considering cultural differences and disabilities.*
- *Encourage testers to think aloud to capture more consistent feedback.*
- *Familiarize ourselves more with WCAG guidelines and use assistive tools for better accessibility testing.*

Compile data

Heuristic			Quantitative		Qualitative:	
Different terms used (e.g., "Capucines") confuse international users.	Search bar lacks history tracking.	Product info inconsistencies confuse users.	Mean task time to find fastest delivery: 90s (3x longer than the optimum 30s).	SUS score: 46, indicating poor usability.	All users expected to sort by price and were frustrated they couldn't.	Users frustrated by difficulty finding product care info.
Hard to find browsing history, adding cognitive load.	Small fonts affect readability, especially on mobile devices.	No customer reviews, affecting trust.	4/6 participants couldn't find product care info, taking too long.	Reading product details took longer, especially due to small fonts.	Most users found small fonts hard to read important info.	Low text contrast made reading difficult, but enhancing contrast solved the issue.
Recently viewed items hard to access.	Menu options too many and repetitive, affecting navigation.	Poor text contrast in images makes it hard to read.				

Organise/structure data

Effectiveness

Different terms used (e.g., "Capucines") confuse international users.

No customer reviews, affecting trust.

Product info inconsistencies confuse users.

Hard to find browsing history, adding cognitive load.

Efficiency

Mean task time to find fastest delivery: 90s (3x longer than the optimum 30s).

4/6 participants couldn't find product care info, taking too long.

Recently viewed items hard to access.

Menu options too many and repetitive, affecting navigation.

Reading product details took longer, especially due to small fonts.

Satisfaction

Small fonts affect readability, especially on mobile devices.

Search bar lacks history tracking.

Low text contrast made reading difficult, but enhancing contrast solved the issue.

All users expected to sort by price and were frustrated they couldn't.

Users frustrated by difficulty finding product care info.

Most users found small fonts hard to read important info.

Conclusion & Insights

1. Difficulty Finding Product Care Information

What you saw/heard:

- Participants **mentioned frustration** with the **"product care"** information being hidden in the bottom navigation bar rather than being integrated into the product details page.

What it means for user behaviour:

- Users who couldn't find this information may **feel uncertain** about the quality and longevity of the products, leading to hesitation in completing their purchase.

Why it matters for usability & user outcomes:

- Difficulty accessing important product information can cause users to **abandon their shopping bags**.

Heuristics

Visibility of system status
(Mental models)

Qualitative data

"I spent a lot of time looking for product care information in the product details but couldn't find it. I had to ask for help."

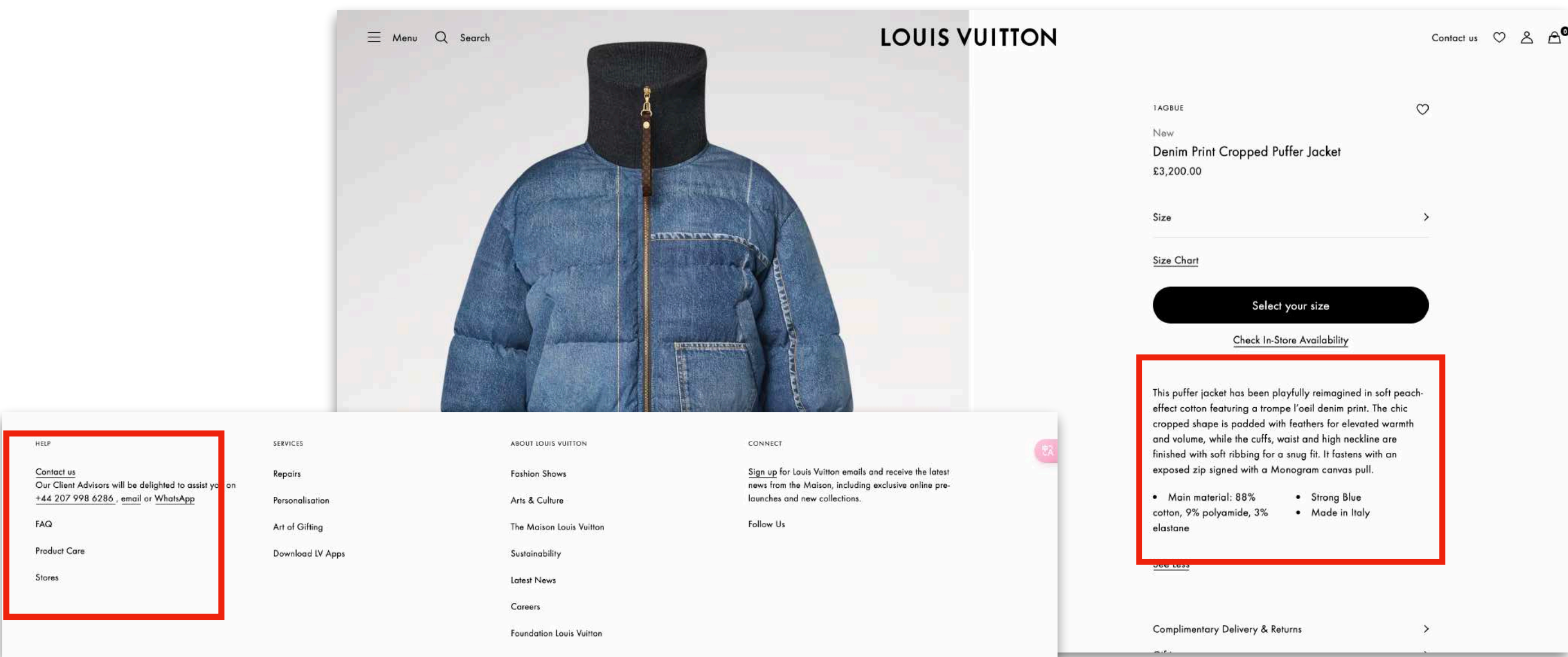
Quantitative data

Efficiency

Task time was longer for users attempting to find this information.

Effectiveness

Many users failed to complete the task (finding product care details), leading to higher error rates.



Conclusion & Insights

2. Filter Function Inconsistency

What you saw/heard:

- Users found that not all filters (like price range) were consistently available across categories. **In some cases, filters were present, while in others, they were missing.**

What it means for user behaviour:

- Inconsistent filtering options **can confuse users.**

Why it matters for usability & user outcomes:

- Inconsistent filtering options disrupt the user's search flow, making it **more difficult to find products.**

Heuristics

Consistency & standards:
(Mental models)

Qualitative data

"It was confusing that sometimes I could filter by price, and other times I couldn't."

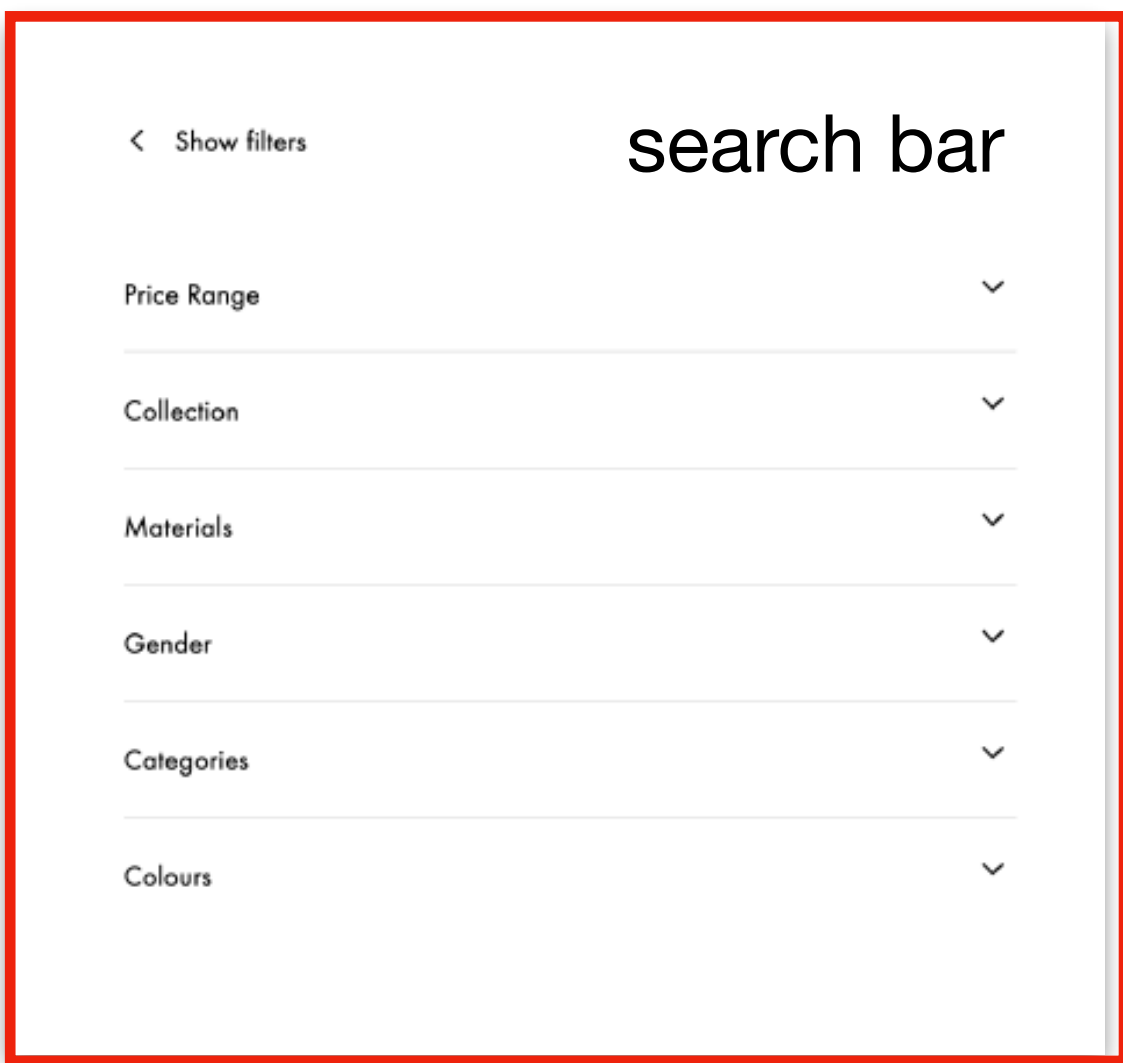
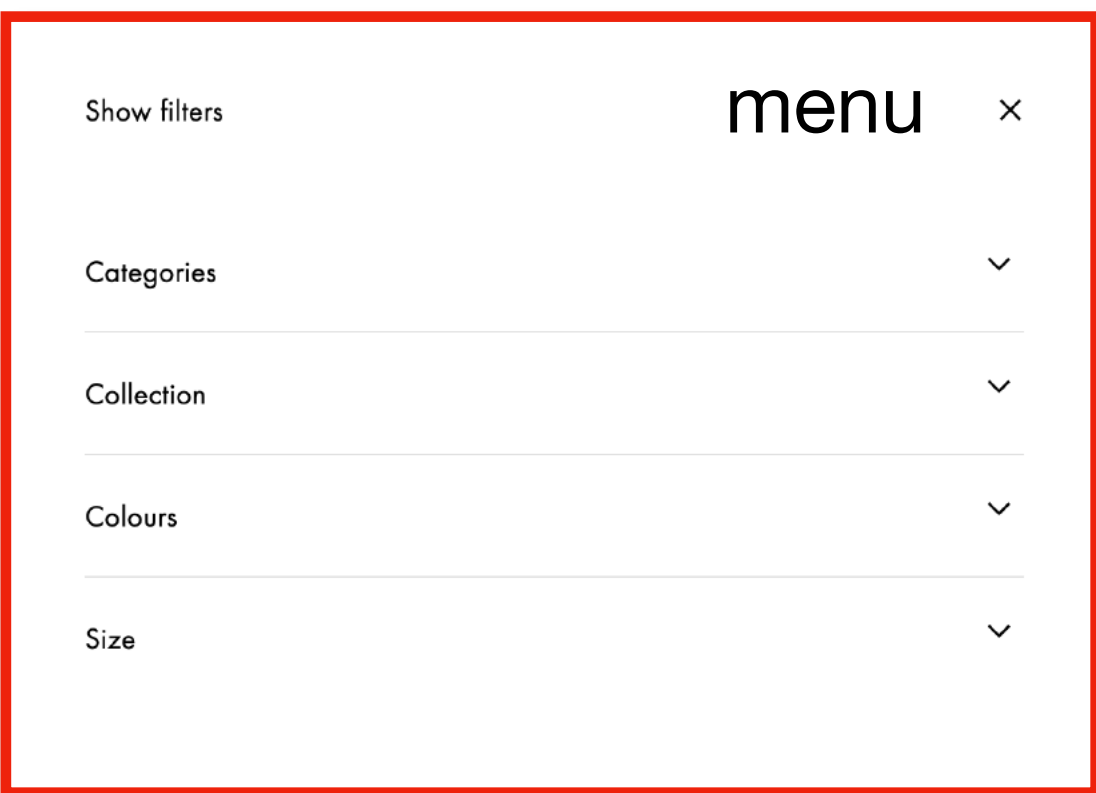
Quantitative data

Efficiency

Increased task time due to users needing to manually sort the prices themselves.

Effectiveness

Higher error rates when users fail to find products they need, due to missing filters.



Conclusion & Insights

3.Small Font Size

What you saw/heard:

- The font size used in some areas, such as product descriptions, **was too small** for easy reading.

What it means for user behaviour:

- Users were forced to zoom in to read, increasing cognitive load.

Why it matters for usability & user outcomes:

- Small text can make a website less accessible, leading to decreased engagement and potential abandonment, **especially for users with visual impairments.**

Heuristics

Aesthetic and minimalist design:
(Gestalt)

Qualitative data

"I couldn't read anything without zooming in.
It was tiring."

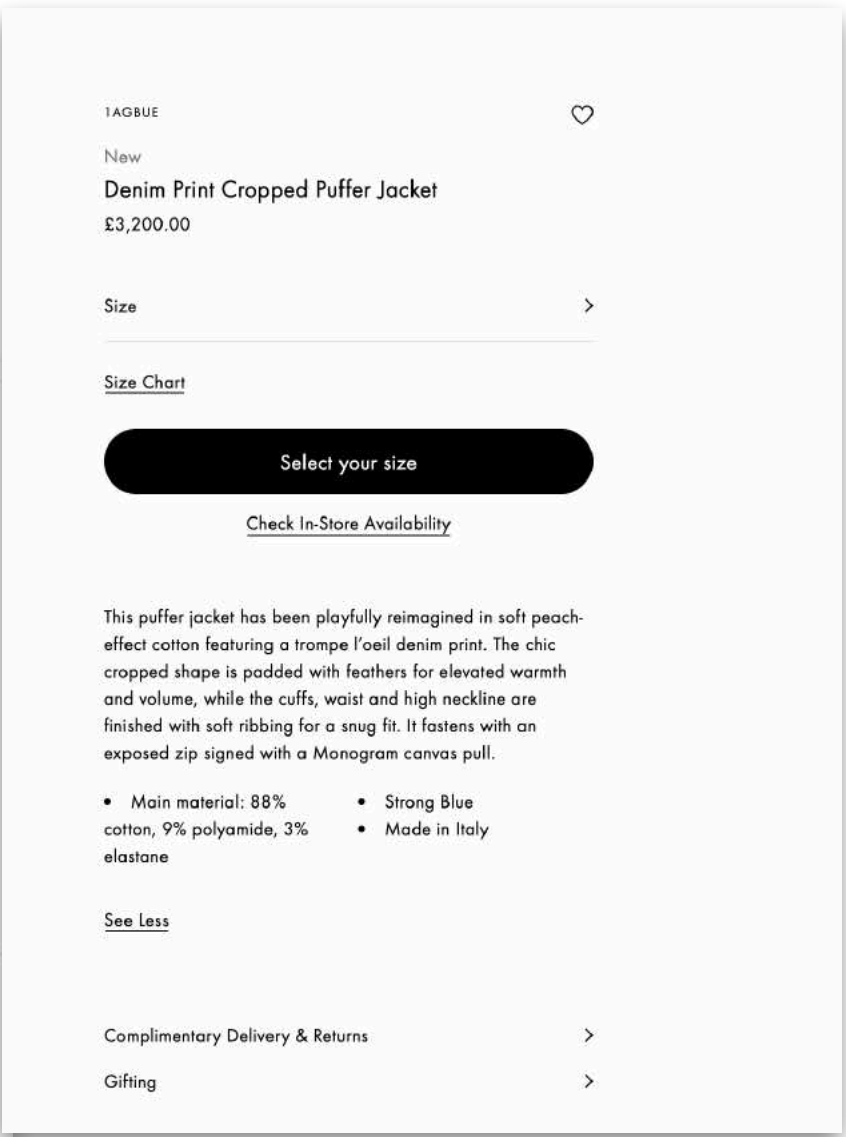
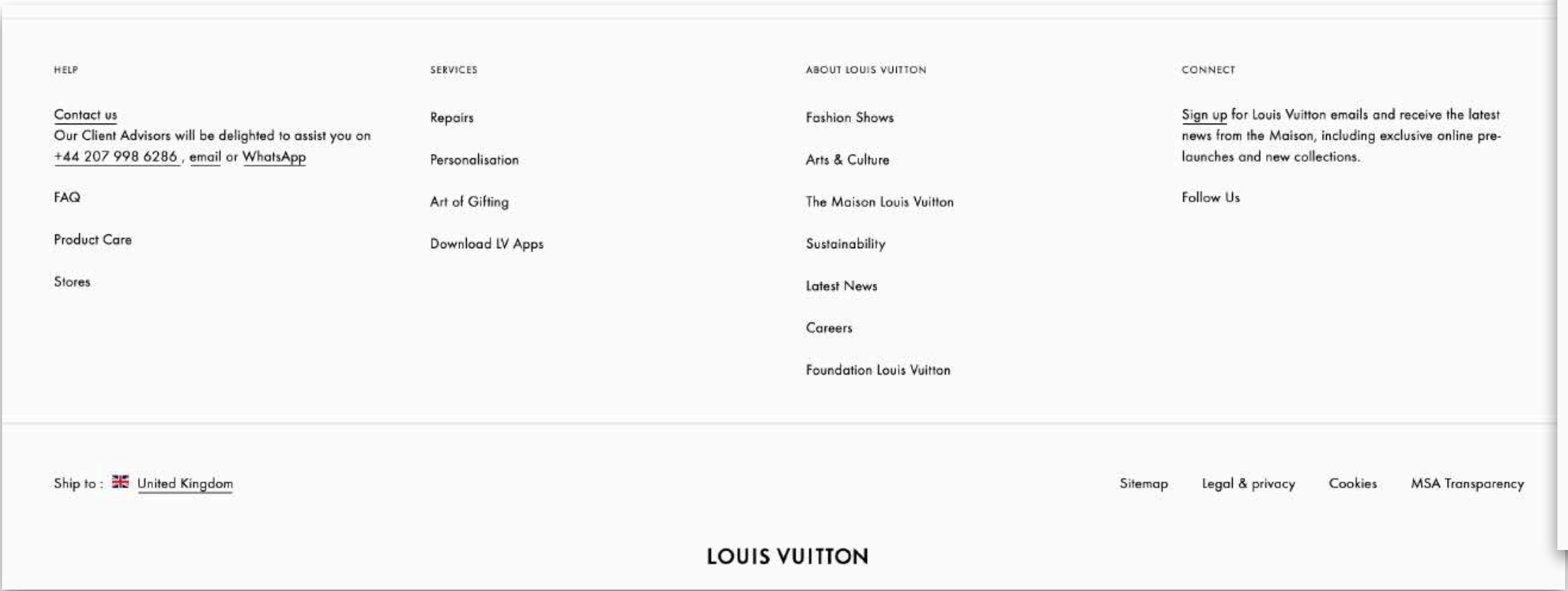
Quantitative data

Efficiency

Task time increased as users had to read the text more carefully.

Satisfaction

Low satisfaction due to difficulties reading product details and information.



Conclusion & Insights

4. Difficult to Find Recently Viewed Products

What you saw/heard:

- Users **had trouble finding recently viewed products**, which required multiple steps to access, often having to go through the account page or product detail pages.

What it means for user behaviour:

- Users cannot easily revisit recently viewed products.

Why it matters for usability & user outcomes:

- Users who want to compare or revisit a product will **be frustrated and give up**.

Heuristics

Recognition rather than recall:

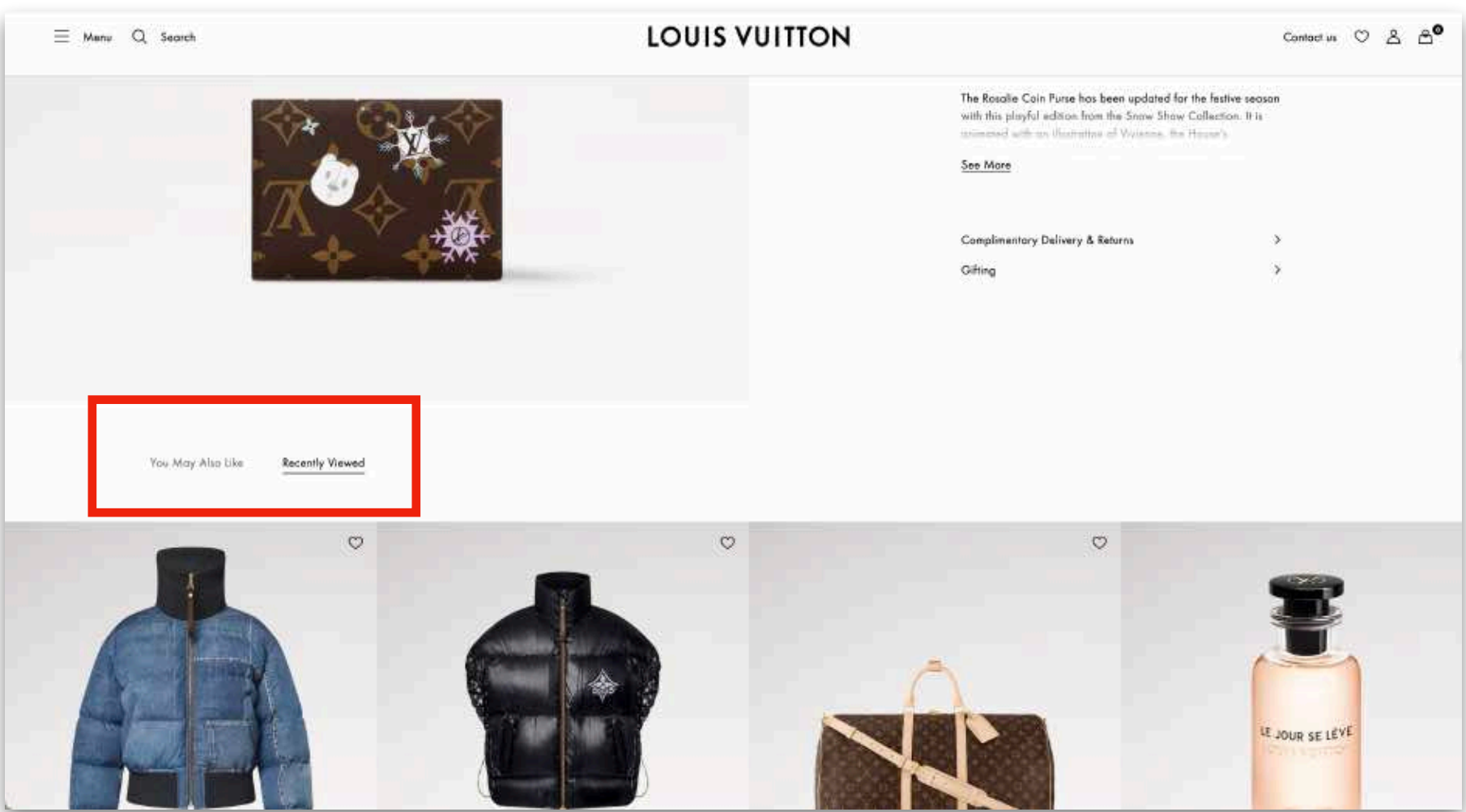
Quantitative data

Effectiveness

Users cannot easily find recently viewed products, resulting in higher error rates on tasks.

Qualitative data

“I can’t find the recently viewed feature”



Conclusion & Insights

5.No Customer Reviews Available

What you saw/heard:

- Users noticed that **there were no customer reviews available** for the products, which is a common feature on most e-commerce sites.

What it means for user behaviour:

- The absence of reviews may **make users feel uncertain** about the quality of the product, leading to reduced confidence in making a purchase.

Why it matters for usability & user outcomes:

- Missing reviews can cause users to give up the website.

Heuristics

Consistency & standards:
(Mental models)

Quantitative data

Satisfaction

SUS score was low.

Qualitative data

"I like to see what other people think before I buy anything. It's frustrating that there are no reviews here."

6.Use Specific Words

What you saw/heard:

- The website uses specific words like "Capucines" without explanation, which **confuses new users**.

What it means for user behaviour:

- Users unfamiliar with luxury brand may **feel alienated or confused**, leading to frustration and hesitation in making purchases..

Why it matters for usability & user outcomes:

- If users cannot easily understand the product name or category, it may **cause them to ignore**

Heuristics

Match between system and the real world
(Mental models)

Quantitative data

Effectiveness

High error rate when users try to search or understand certain words,

Qualitative data

"I had no idea what 'Capucines' meant. It sounds fancy"

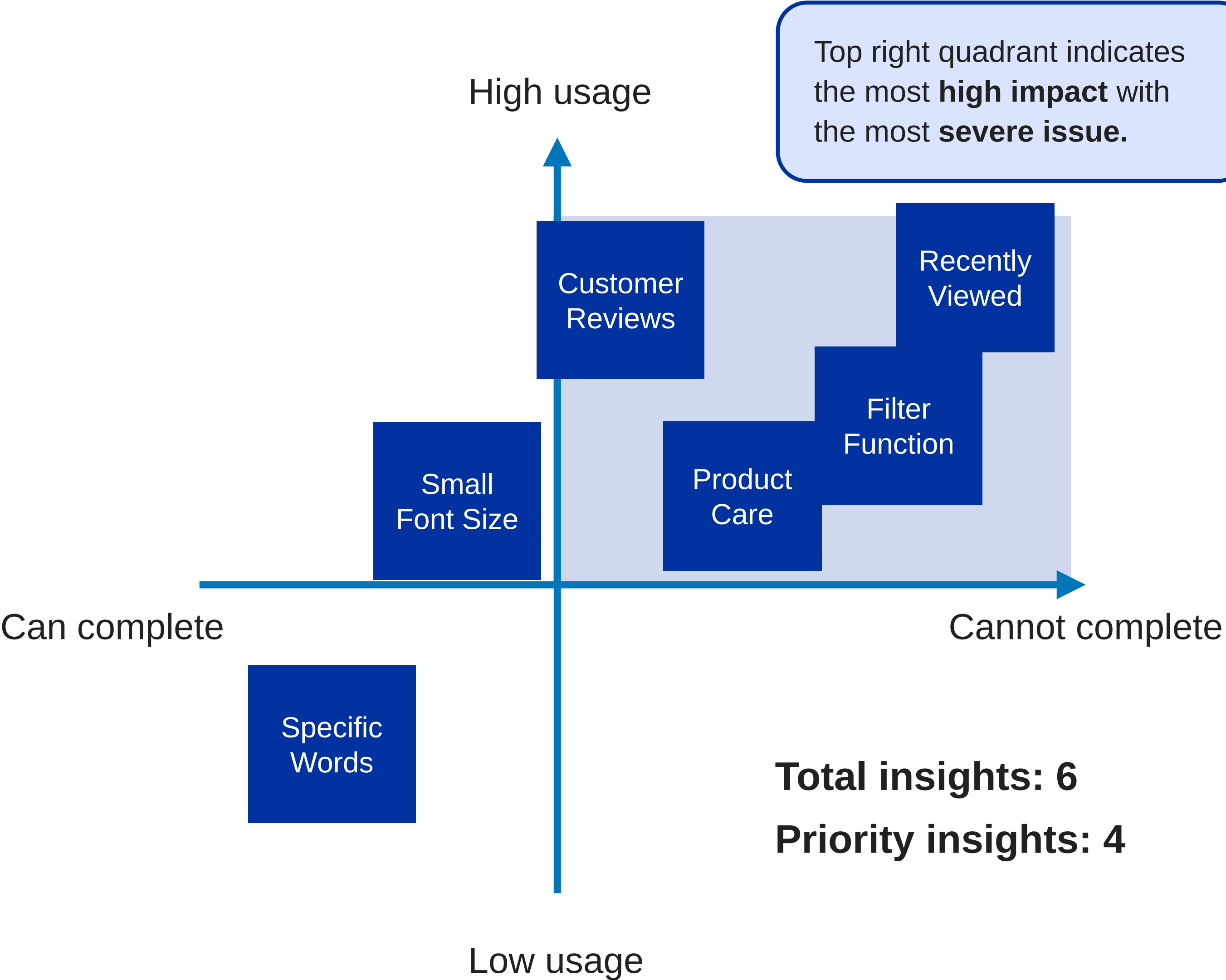
Insight Prioritisation

Prioritisation Matrix

Axes:

- 1. **Users Frequency of usage** - high impact feature or flow
- 2. **Ability to complete goal** - How severe the issue was

Difficulty Finding Product Care Information
Filter Function Inconsistency
Difficult to Find Recently Viewed Products
No Customer Reviews Available



Redesign Home Page

Increased Font Size

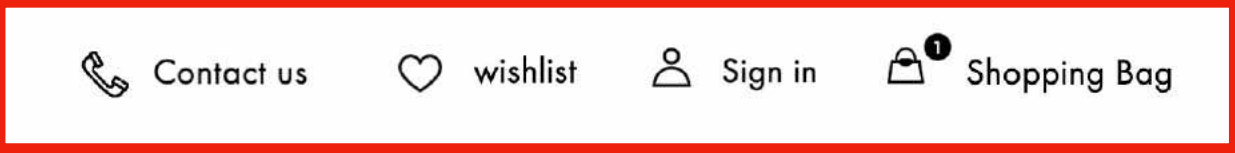


Font size has been increased to improve readability, reducing strain on users and making it easier for them to consume information without zooming in.

Heuristic - Aesthetic and Minimalist Design
Psych - Mental Models, Gestalt



LOUIS VUITTON



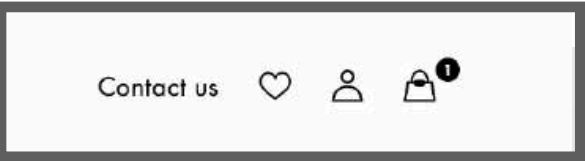
Added Matching Icons and Text in Menu



Added icons to the menu to match the text, helping users quickly identify and understand each category or option.

Heuristic - Recognition rather than recall
Psych - Dual Coding Theory

original



Close

- Women
 - Men
 - Travel
 - Collections
- Bags
 - Shoes
 - Accessories
 - Clothing
 - Fashion Jewellery and Watches

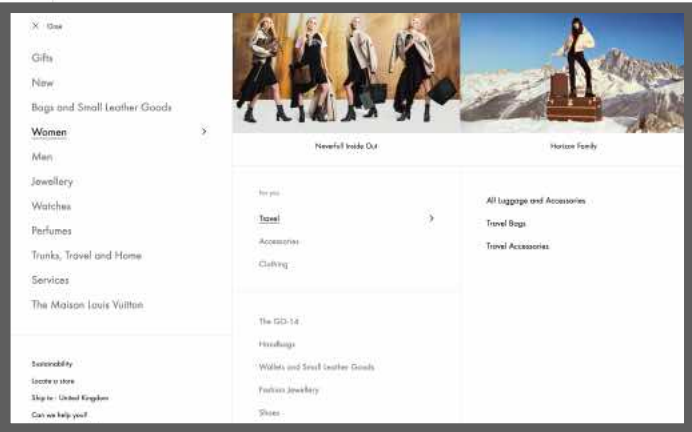
Reorganize Menu Structure



The original menu layout was too complex, and the categories were reorganized to be clearer and similar items were merged.

Heuristic - Recognition rather than recall
Psych - Mental Models

original



Added Image Matching



Adding images makes it easier to visually identify product categories

Heuristic - Match between system and real world
Psych - Dual Coding Theory

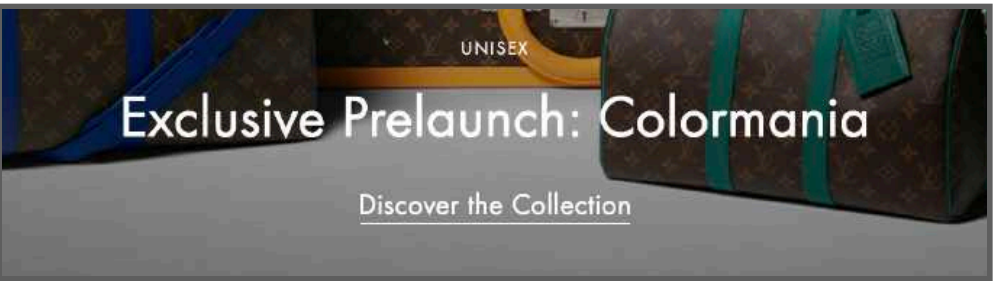


Increased Text Contrast



Increase contrast between text and images to improve readability and ensure text is more legible against the background.
Heuristic - Accessibility-1.4.3 Contrast (Minimum)
Psych - Visual Perception

original



Tags for Usability Issues

- Psychology Principles
- Qualitative Testing
- Heuristic Evaluation
- Accessibility Evaluation
- Quantitative Testing
- Insights from Testing

Redesign Product Listing Page

Added Back Button



A back button was missing from the original website design and was added to help users return to a previous page.

Heuristic - User Control and Freedom

Psych - Mental Models

Added Page Titled



Page titles are necessary to help users identify their location within a website. They are also necessary under Guideline 2.4.2 of WCAG 2.2.

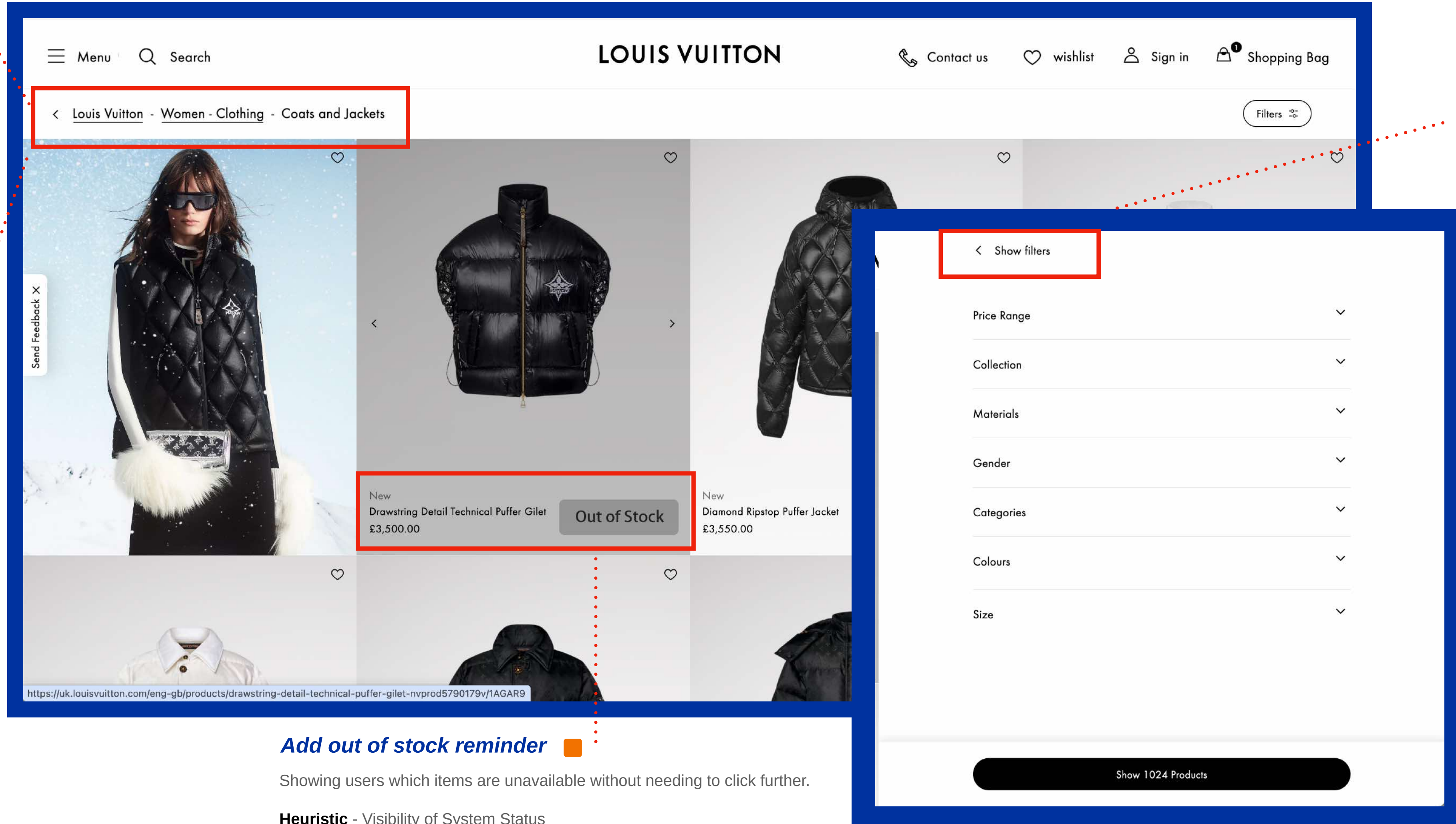
Titles are written in capital letters to echo the aesthetic of the original design

Heuristic - User Control and Freedom

Psych - Mental Models

Tags for Usability Issues

- Psychology Principles
- Heuristic Evaluation
- Quantitative Testing
- Qualitative Testing
- Accessibility Evaluation
- Insights from Testing



Unified Filter System

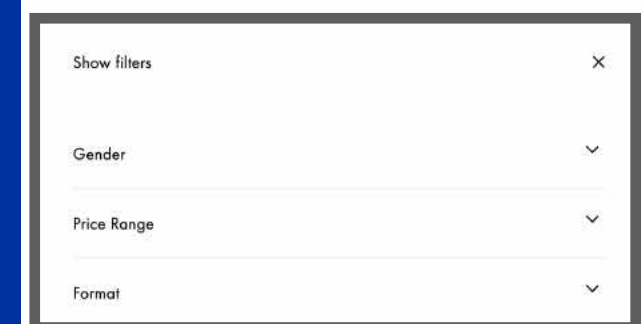


The filters on the website are always different, which confuses users. Unified filtering system allows users to experience a consistent filtering system when browsing all product categories.

Heuristic - consistency and standards

Psych - mental models

original



Add out of stock reminder

Showing users which items are unavailable without needing to click further.

Heuristic - Visibility of System Status

Psych - Recognition Over Recall

Redesign Product Page

Added Back Button



A back button was missing from the original website design and was added to help users return to a previous page.

Heuristic - User Control and Freedom

Psych - Mental Models

Added Page Titled



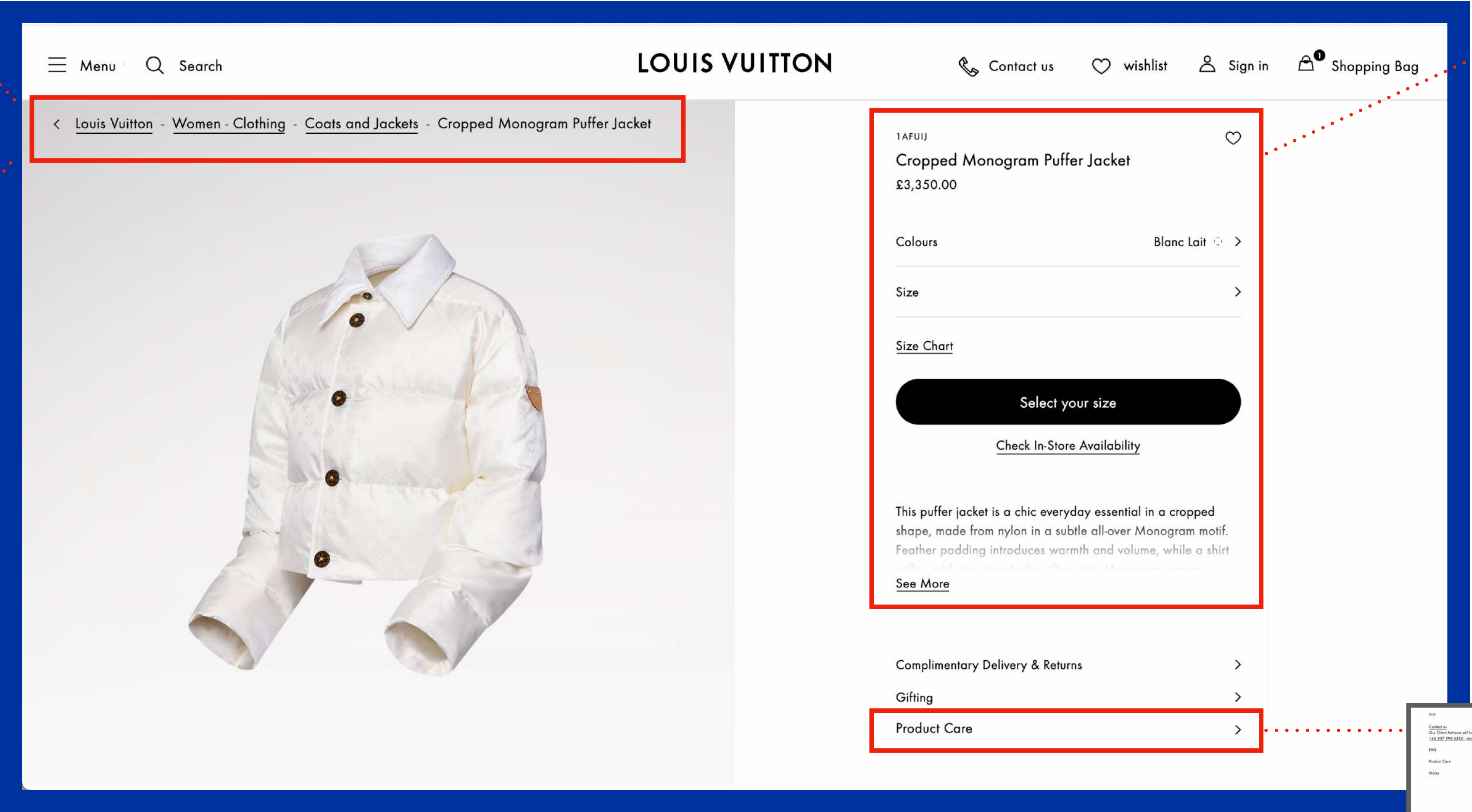
Page titles are necessary to help users identify their location within a website. They are also necessary under Guideline 2.4.2 of WCAG 2.2. Titles are written in capital letters to echo the aesthetic of the original design

Heuristic - User Control and Freedom

Psych - Mental Models

Tags for Usability Issues

- Psychology Principles
- Heuristic Evaluation
- Quantitative Testing
- Qualitative Testing
- Accessibility Evaluation
- Insights from Testing



Increased Font Size



Font size has been increased to improve readability, reducing strain on users and making it easier for them to consume information without zooming in.

Heuristic - Aesthetic and Minimalist Design

Psych - Mental Models, Gestalt

Relocated Product Care Information

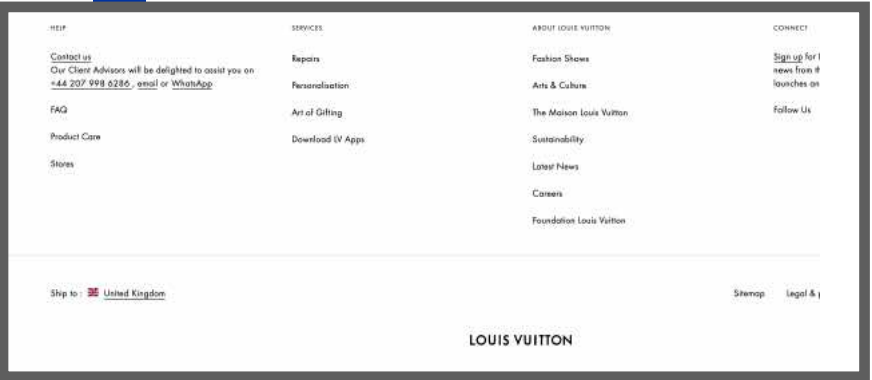


Moving the product care section to the product details area makes the details easy to find without additional navigation.

Heuristic - Consistency and Standards

Psych - Mental Models

original



Thank You!